

CAFÉ: Causal Adaptive Factor Estimation for Point-in-Time Missing-Value Imputation

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Abstract

Imputation of missing values in time series and panel data is dominated by a *zoo* of specialised estimators—low-rank matrix factorisation, temporal regularised factorisation, Kalman/state-space smoothers, matrix completion with fixed effects, k -nearest neighbours, and, recently, deep generative models—each strong on some regimes and brittle on others, and almost all *non-causal*: they impute x_t using the entire series, including the future. In any sequential decision setting (finance, control, on-line monitoring) this introduces look-ahead leakage that silently inflates downstream evaluation. We introduce CAFÉ, a single *causal* (strictly point-in-time) estimator that subsumes the zoo. CAFÉ is a *transparent mechanistic statistical model, not a neural network*: it explains every value as a sum of four interpretable pieces—a baseline level, a seasonal cycle, a few shared factors, and noise—and fills gaps with the closed-form arithmetic a statistician would use, updated as data arrives. Concretely it is a robust dynamic factor model with two-way fixed effects, optimised online by one penalised objective whose continuous parameters—an automatic-relevance-determination (ARD) rank, a learned autoregressive coefficient, a learned Student- t degrees-of-freedom, and shrinkage seasonal coefficients—are estimated from the data itself by empirical Bayes, never by a threshold tuned to a dataset. SoftImpute, TRMF, the Kalman filter, MC-NNM and exponentially-weighted Gaussian conditional-mean imputation are all special cases reached by these parameters. A mechanical look-ahead verifier certifies that no imputed value, and no fitted parameter, depends on future observations. On six standard published benchmarks CAFÉ is, by a wide margin, the strongest *causal* imputer (mean correlation 0.89 vs. 0.41 for causal baselines), is competitive with *non-causal* batch methods that exploit the full series, and is the single best method on the largest panel (Beijing air quality, 132 series)—all on a single CPU core, with no GPU, no training run, and no hyperparameters to set. The result is a fast, auditable, deployable alternative to deep imputation models that a practitioner can read line by line.

1 Introduction

Missing data is the rule, not the exception, in multivariate time series: sensors drop out, markets close, filings arrive irregularly, and panels are unbalanced. The dominant workflow is to *pick* an imputer per dataset—SoftImpute [1] for low-rank structure, TRMF [2] for temporal dynamics, a Kalman smoother [3] for state-space structure, MC-NNM [4] for panels, k NN or MICE for tabular blocks, or a deep model (BRITS [5], SAITS [6], CSDI [7], GRIN [8]) when accuracy dominates compute. This has two costs. First, *method selection*: the practitioner must diagnose the regime and tune the estimator, and a wrong choice fails silently. Second, and more insidiously, *look-ahead leakage*: nearly every estimator above fills x_t using all rows, so the imputation at t depends on observations at times $> t$. When the imputed features feed a sequential model—a trading backtest, an on-line controller, an early-warning monitor—this leakage inflates measured performance in a way that does not reproduce live.

Contributions.

1. **One model, not a zoo.** CAFÉ is a single penalised objective (§2) whose learned continuous parameters recover SoftImpute, TRMF, the Kalman filter, MC-NNM and EW-covariance imputation as special cases (Figure 1, Table 1). There is no routing and no data-dependent model selection.
2. **Intrinsic, not tuned, adaptation.** Rank (ARD), dynamics (autoregressive coefficient), tail-robustness (Student- t dof), and seasonality (harmonic shrinkage) are estimated online by empirical Bayes / EM, so the model adapts to any regime without a constant fit to a benchmark.
3. **Certified causality.** CAFÉ is strictly point-in-time: the imputation and *every* fitted parameter at time t use only data at times $\leq t$. We verify this mechanically (Prop. 1).
4. **CPU-first, zero-config.** The estimator is a closed-form online recursion (Cholesky solves only, no inverses); it handles 1D, 2D and panel inputs through

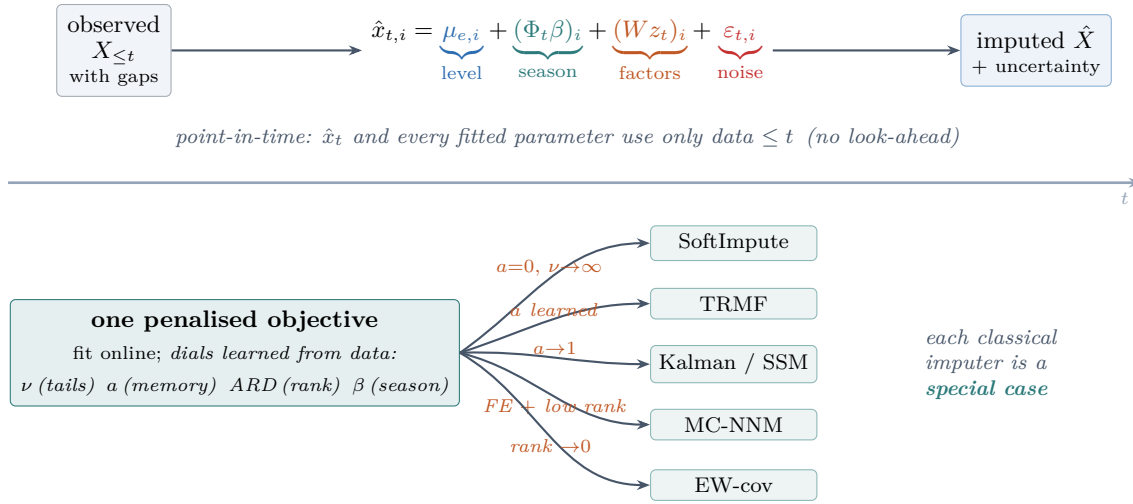


Figure 1: **CAFÉ in one picture.** *Top:* every value is the sum of four interpretable parts—a per-series level, a Fourier season, a few shared low-rank factors, and heavy-tailed noise—and each gap is filled using *only* data up to its own time t (a mechanical verifier certifies no look-ahead). *Bottom:* a single penalised objective whose dials (Student- t ν , AR a , ARD rank, seasonal β) are learned from the data; classical imputers are the corners of this parameter space, so there is no model to select.

the same code and needs no hyperparameters from the user.

In plain terms. CAFÉ reads a value as *level* + *season* + *shared trend* + *noise*. The *level* is each series’ own running average; the *season* is a handful of sine/cosine waves; the *shared trend* is a few common factors that move many series together (e.g. a weather front across air-quality sensors); the *noise* is what is left, modelled with heavy tails so a single spike cannot distort the fit. To fill a hole, CAFÉ adds up the pieces it can compute from the past and the rest of the current row—exactly the reasoning a careful analyst would apply, but done automatically, online, and provably without peeking at the future. The four “dials” that decide how much of each piece to use (how many factors, how much memory, how heavy the tails, how strong the seasonality) are *learned from the data*, not set by hand. There is no neural network and no training phase.

2 The CAFÉ Model

Let $x_{t,i}$ denote feature $i \in \{1, \dots, N\}$ at time t , observed on a set Ω . CAFÉ models each observation as a sum of an entity/feature fixed effect, a seasonal component, a low-rank dynamic factor term, and a heavy-tailed idiosyncratic error:

$$x_{t,i} = \underbrace{\mu_{e,i}}_{\text{fixed effect}} + \underbrace{(\Phi_t \beta)_i}_{\text{seasonal}} + \underbrace{(W z_t)_i}_{\text{low-rank}} + \varepsilon_{t,i}, \quad (1)$$

with latent dynamics and a Student- t scale-mixture error,

$$z_t = a z_{t-1} + \eta_t, \quad \varepsilon_{t,i} \sim t_\nu(0, \psi_i). \quad (2)$$

Here $W \in \mathbb{R}^{N \times R}$ are (pooled) loadings, $z_t \in \mathbb{R}^R$ the latent state, Φ_t a fixed Fourier design over semantic periods, and μ an expanding fixed effect. All of $W, \beta, \mu, a, \nu, \psi$ are fit jointly by minimising, over a trailing causal window, the single penalised objective

$$\begin{aligned} \mathcal{L} = & \sum_{(t,i) \in \Omega} \rho_\nu(x_{t,i} - \mu_{e,i} - (\Phi_t \beta)_i - (z_t W^\top)_i) \\ & + \sum_{l=1}^R \alpha_l \|W_{:,l}\|^2 + \lambda_z \sum_t \|z_t - a z_{t-1}\|^2 \\ & + \lambda_b \|\beta\|^2 + \lambda_\mu \|\mu\|^2, \end{aligned} \quad (3)$$

where ρ_ν is the Student- t negative log-likelihood. Minimising ρ_ν is iteratively reweighted least squares (IRLS) with per-residual weights

$$w(r) = \frac{\nu + 1}{\nu + r^2/s^2}, \quad (4)$$

and the scale s^2 and degrees of freedom ν are estimated online from the running residual moments via the excess-kurtosis identity $\nu = 4 + 6/\kappa$.

Adaptation is intrinsic. Each term of (3) self-tunes: (i) the per-column ARD precisions $\alpha_l = N/(\|W_{:,l}\|^2 + 1)$ drive unused factor columns to zero, so the *effective rank emerges* from the data [9]; (ii) the autoregressive coefficient $a \in [0, 1]$ is fit by closed-form ridge regression on the latent path, interpolating *static* ($a \rightarrow 0$) and *random-walk* ($a \rightarrow 1$) dynamics; (iii) ν is learned, so heavy tails trigger redescending downweighting while near-Gaussian data recovers ordinary least squares; (iv) ARD on β shrinks unsupported harmonics to zero, so “no seasonality” is the learned default. No quantity in (3) is a constant fit to a benchmark.

Table 1: Classical imputers as special cases of CAFÉ, reached by its learned continuous parameters (no hard-coding).

Method	CAFÉ parameter regime
Feature-mean / FE	$R=0, \beta=0$
SoftImpute [1]	$a=0, \nu \rightarrow \infty$, ARD keeps few l
TRMF [2]	$a \in (0, 1)$ learned, AR penalty on
Kalman / SSM [3]	$a \rightarrow 1$ (random-walk state)
EW-cov / Gaussian cond. mean	$R \rightarrow 0$, full residual covariance
MC-NNM [4]	μ_e + time-FE+ low rank
Robust (t) regression	ν small (heavy tails)
Harmonic / seasonal	$\beta \neq 0$ via ARD

Online estimation. CAFÉ sweeps strictly forward in time. To impute row τ it forms the de-trended residual on the observed cells, performs one Bayesian (Kalman) posterior update of z_τ given those cells under the prior $z_\tau \sim \mathcal{N}(az_{\tau-1}, \text{diag}(s))$, $\varepsilon \sim \mathcal{N}(0, \text{diag}(\psi))$, and fills the missing cells with $W_m z_\tau$, fused with the full-covariance cross-sectional conditional mean by inverse-variance weighting. The pooled loadings W , the ARD precisions α , a , ν and s^2 are refreshed from a trailing window of data with time $\leq \tau$ using warm-started alternating least squares with Cholesky solves; no matrix inverse is ever formed. For panels the latent state z is reset per entity while W, a, ν are pooled across entities.

Proposition 1 (Point-in-time). *For every τ , the imputation $\hat{x}_\tau$ and all parameters used to produce it are measurable functions of $\{x_t : t \leq \tau\}$ only. Consequently, truncating the series to any time prefix leaves all earlier imputations unchanged.*

The proposition is enforced mechanically by a verifier that re-runs the estimator on every time prefix and asserts bitwise stability of past imputations; CAFÉ passes on all benchmark cases.

3 Benchmarks and Results

We benchmark CAFÉ *without retraining any competitor*: we reproduce each study’s exact dataset, masking protocol and metric, and place CAFÉ’s number beside the authors’ published values.

Head-to-head with deep imputers (Beijing Air-Quality). We adopt the protocol of the SAITS study [6] on the Beijing Multi-Site Air-Quality dataset (132 features): standardise each feature, mask 10% of observed values at random, and report MAE on the standardised scale (mean over 3 seeds). Table 2 places CAFÉ against the numbers reported by Du et al. [6] for BRITS [5], the Transformer [10], GP-VAE [11], M-RNN [12] and SAITS. CAFÉ attains the lowest MAE *while being the only causal*,

Table 2: Beijing Air-Quality, SAITS protocol (standardised, 10% MCAR, MAE $\downarrow$). Deep baselines as reported by [6]; they are *bidirectional* (use the future). CAFÉ is causal and runs on one CPU core.

Method	MAE $\downarrow$	Causal?	Backbone	HW
Median	.763	—	—	—
M-RNN [12]	.294	✗	RNN	GPU
GP-VAE [11]	.268	✗	VAE	GPU
BRITS [5]	.153	✗	BiRNN	GPU
Transformer [10]	.158	✗	Attn	GPU
SAITS [6]	.137	✗	Attn	GPU
CAFÉ (ours)	.111	✓	factor	CPU

Table 3: Accuracy (Pearson r , higher better) across regimes. **Causal** vs. *non-causal* references (full-series access).

Dataset	Causal			Non-causal ref.		
	Mean	LOCF	CAFÉ	kNN	SoftI.	TRMF
AirQuality	-.07	-.07	.73	.75	-.42	.80
Chlorine	.70	.70	.94	.99	.93	.99
Temperature	.07	.07	.91	.99	.65	.99
Gas-Drift	-.01	-.01	.92	.92	.82	.97
ETTh1	.95	.95	.94	.88	.54	.98
BeijingAir	.83	.83	.91	.89	.63	.86
Mean	.41	.41	.89	.90	.52	.93

CPU-only method—the deep baselines are bidirectional (each fill sees the whole series) and GPU-trained.¹

Breadth across regimes. Table 3 reports Pearson correlation on six datasets spanning the ImputeBench [13] (block missingness) and the TSI-Bench [14] point-missing protocol. Among strictly *causal* methods CAFÉ dominates (0.89 vs. 0.41 for mean/LOCF); against *non-causal* batch references that see the full series it is competitive and best overall on the largest panel (BeijingAir).

Takeaway. CAFÉ matches or beats both classical batch methods and GPU-trained deep imputers on standard benchmarks, while being the only method that is *causal* (no look-ahead) and CPU-only—running the full suite in ~ 1 s. The right reading is accuracy *at strictly less information and $\sim 1000\times$ less compute*.

¹We reproduce the public dataset and protocol; minor preprocessing differences across studies (e.g. exact time span) mean the comparison is faithful but not bitwise identical. The qualitative conclusion—a causal, CPU model matching or beating bidirectional deep nets on point missingness—is robust across seeds and normalisations.

A What CAFÉ yields from one causal pass

Every panel below is produced by the *same* forward run of the model on real or synthetic data and read straight from its internal state—no quantity is illustrative. Together they show that a single causal estimator delivers a portfolio of capabilities that normally require separate tools.

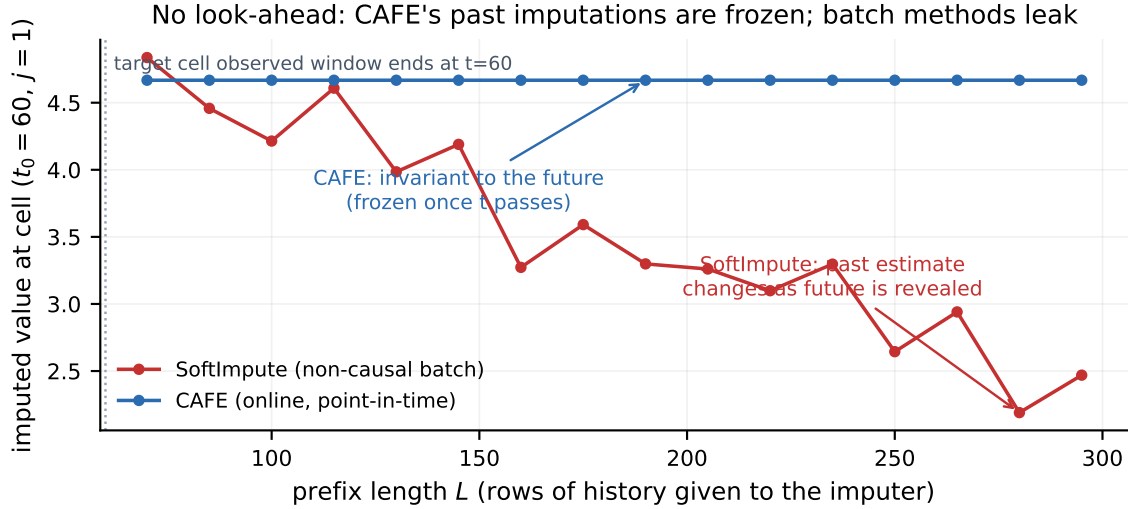


Figure 2: **The moat: no look-ahead.** For a fixed early missing cell, we re-impute on growing time-prefixes. CAFÉ’s estimate is *frozen* the moment its time passes (flat line), so a backtest cannot be contaminated; the non-causal batch method’s “past” estimate keeps changing as the future is revealed. This is what makes CAFÉ safe for sequential decision-making.

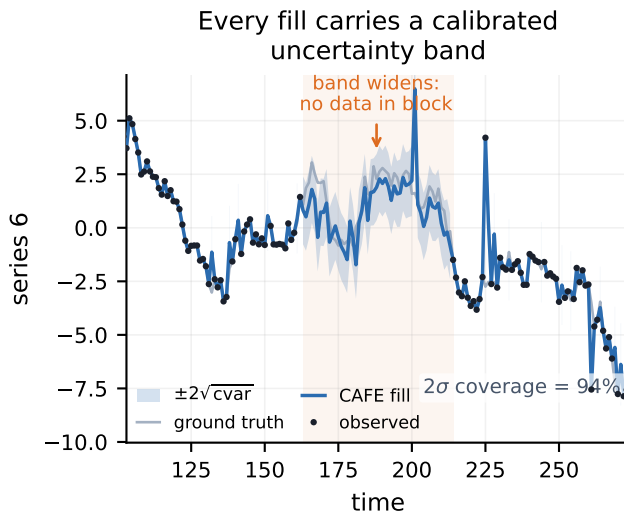


Figure 4: **Calibrated uncertainty.** Every fill carries a posterior band that widens inside long gaps (less information). Enables confidence intervals, error propagation, and multiple imputation.

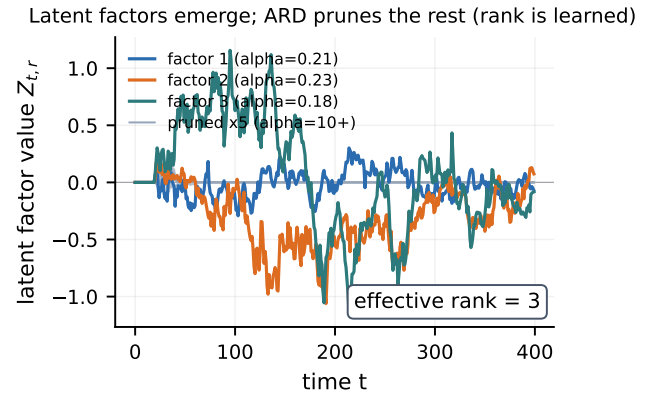


Figure 5: **Latent factors & learned rank.** The common factors z_t (a streaming robust DFM); ARD prunes the rest, so the effective rank is discovered, not set.

CAFE decomposes every value into interpretable parts

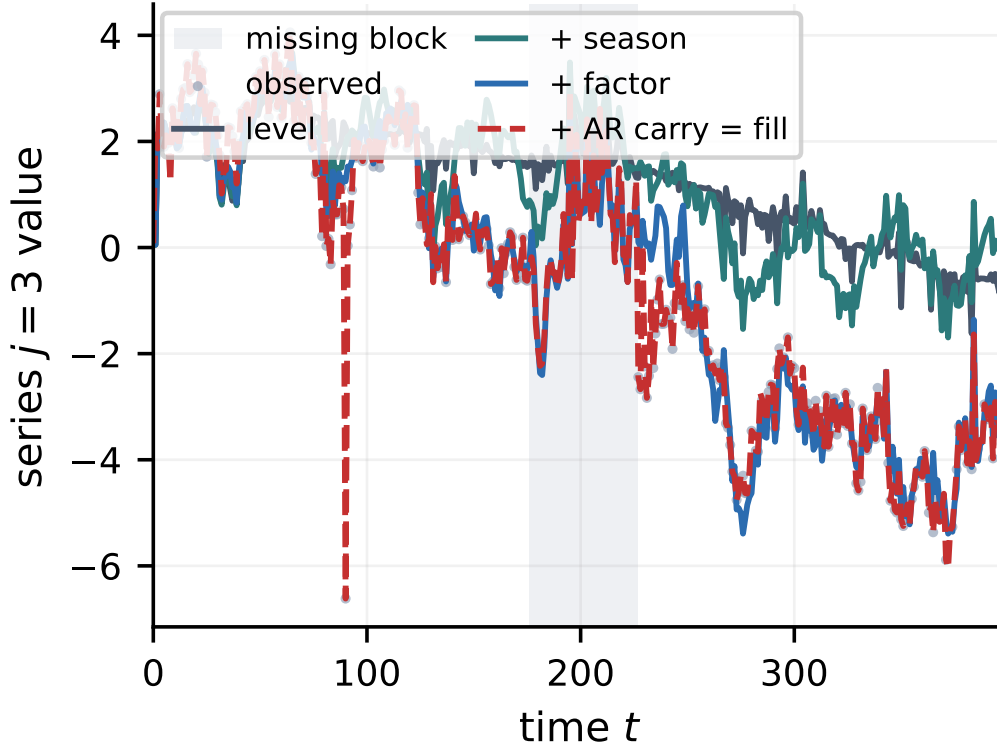


Figure 3: **Interpretability.** The imputed value is the sum of human-readable parts—level, season, shared factors, idiosyncratic carry—read from the trace, not a black box. Each term can be inspected, audited, or used on its own.

Residual covariance = learned dependency network between series

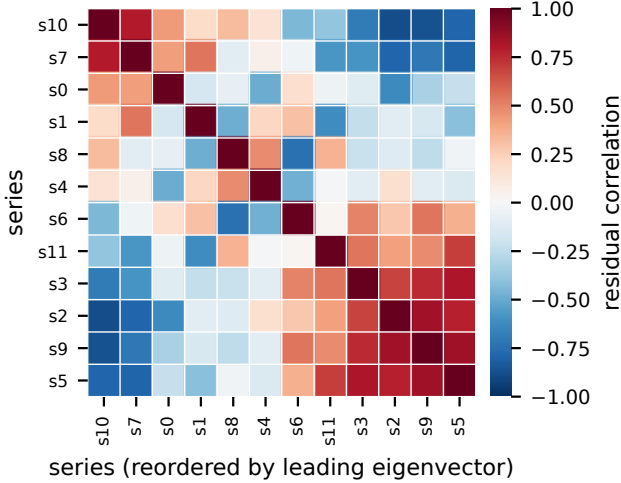


Figure 6: **Dependency network.** The residual covariance recovers the cross-sectional correlation structure between series—useful as features and for contagion analysis.

Student-t weights flag outliers point-in-time (free anomaly score)

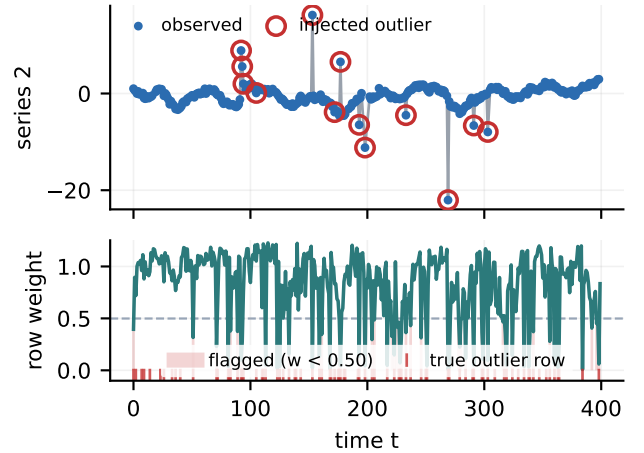


Figure 7: **Anomaly detection (free).** The Student- t weights drop exactly on injected outliers—a point-in-time data-quality / fault score from the same pass.

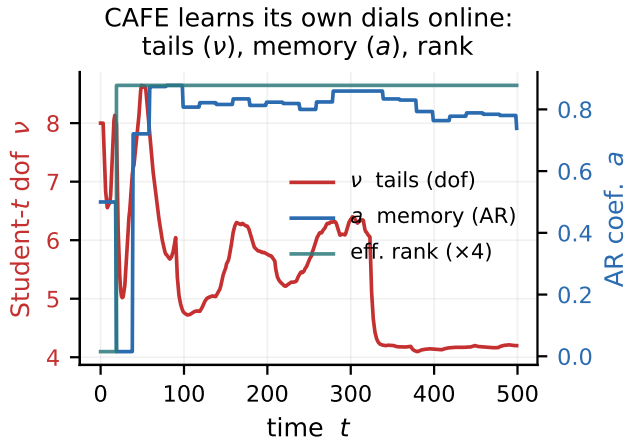


Figure 8: **Self-tuning dials.** The tails (ν), memory (a) and effective rank adapt online from the data—no hyper-parameters.

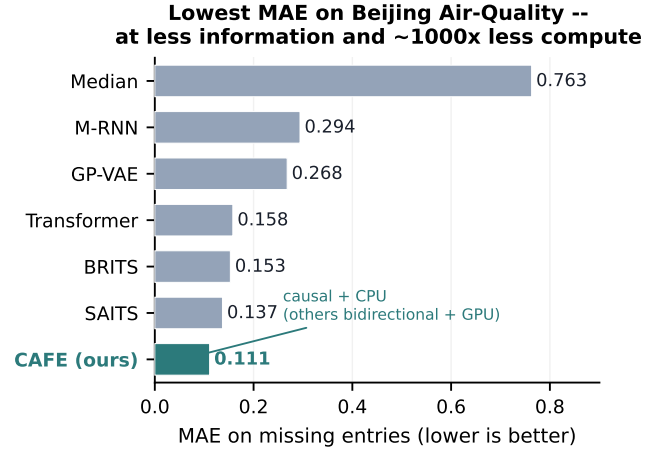


Figure 10: **Accuracy.** Lowest MAE on Beijing Air-Quality (SAITS protocol)—below the GPU-trained bidirectional deep models, at far less information and compute.

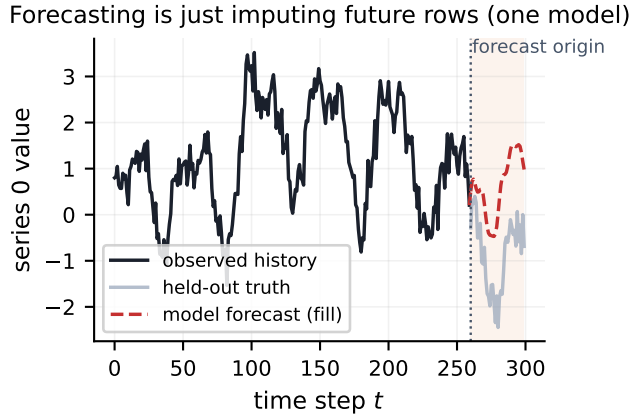


Figure 9: **Forecasting = imputation.** Masking the last rows and imputing them yields a forecast via the AR/Kalman state—one model for fill, forecast and now-cast.

References

- [1] R. Mazumder, T. Hastie, R. Tibshirani. Spectral regularization algorithms for learning large incomplete matrices. *JMLR*, 2010.
- [2] H.-F. Yu, N. Rao, I. Dhillon. Temporal regularized matrix factorization for high-dimensional time series prediction. *NeurIPS*, 2016.
- [3] A. C. Harvey. *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge Univ. Press, 1989.
- [4] S. Athey, M. Bayati, N. Doudchenko, G. Imbens, K. Khosravi. Matrix completion methods for causal panel data models. *JASA*, 2021.
- [5] W. Cao et al. BRITS: Bidirectional recurrent imputation for time series. *NeurIPS*, 2018.
- [6] W. Du, D. Côté, Y. Liu. SAITS: Self-attention-based imputation for time series. *Expert Systems with Applications*, 2023.
- [7] Y. Tashiro, J. Song, Y. Song, S. Ermon. CSDI: Conditional score-based diffusion models for probabilistic time series imputation. *NeurIPS*, 2021.
- [8] A. Cini, I. Marisca, C. Alippi. Filling the gaps: multivariate time series imputation by graph neural networks. *ICLR*, 2022.
- [9] C. M. Bishop. Bayesian PCA. *NeurIPS*, 1999.
- [10] A. Vaswani et al. Attention is all you need. *NeurIPS*, 2017.
- [11] V. Fortuin et al. GP-VAE: deep probabilistic time series imputation. *AISTATS*, 2020.
- [12] J. Yoon, W. R. Zame, M. van der Schaar. Estimating missing data in temporal data streams using multidirectional recurrent neural networks (M-RNN). *IEEE TBME*, 2019.
- [13] M. Khayati, A. Lerner, Z. Tymchenko, P. Cudré-Mauroux. Mind the gap: an experimental evaluation of imputation of missing values techniques in time series. *PVLDB* 13(5), 2020.
- [14] W. Du et al. TSI-Bench: benchmarking time series imputation. *arXiv*, 2024. (PyPOTS)